

Generative Models in Finance

Week 2: Reinforcement Learning Training of LLMs

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Overview

- ➊ From Pre-training to Fine-Tuning
- ➋ Reinforcement Learning Foundations for LLMs
- ➌ RLHF: PPO, GRPO, and the Training Pipeline
- ➍ Knowledge Distillation
- ➎ RL for Mathematical Reasoning

Reference: R. Patel, *Understanding Reinforcement Learning for Model Training, and Future Directions with GRAPE*, [references/llm_training.pdf](#), 2025.

Part 1: From Pre-training to Fine-Tuning

- Recall the standard LLM training pipeline:
 - ① **Pre-training**: next-token prediction on a large text corpus, yielding a base model π_{base}
 - ② **Supervised Fine-Tuning (SFT)**: adapt the base model on curated (prompt, response) pairs to produce π_{SFT}
 - ③ **RLHF / Preference Alignment**: further optimise π_{SFT} using human (or AI) preference feedback
- We will cover all three stages: **pre-training, fine-tuning, and alignment via reinforcement learning**
- Throughout, we denote the policy (i.e. the language model) by π_{θ} , parameterised by $\theta \in \mathbb{R}^d$

What is Pre-training?

- **Pre-training** is the first and most expensive stage of LLM development
- The model learns from a massive corpus of text in a **self-supervised** fashion: no human labels are required
- The learning signal comes from the data itself — specifically, from the task of **next-token prediction**:

Given context $(x_1, \dots, x_{t-1})$, predict x_t

- The resulting model π_{base} is a **base model**: it models $P(x_t \mid x_{<t})$ and can sample continuations, but it has not been trained to condition on instructions or produce structured responses
- Pre-training determines the support of the learned distribution; it encodes the statistical regularities of the training corpus
- **Scale**: modern base models are trained on $\sim 10^{13}$ tokens using $\sim 10^4$ GPUs for $\sim 10^{24}$ FLOPs

The Pre-training Objective

- The pre-training loss (Radford et al., 2018) is the **cross-entropy** (equivalently, negative log-likelihood) over the training corpus $\mathcal{C} = (x_1, x_2, \dots, x_N)$:

$$\mathcal{L}_{\text{PT}}(\theta) = -\frac{1}{N} \sum_{t=1}^N \log \pi_{\theta}(x_t \mid x_1, \dots, x_{t-1}) \quad (2.1)$$

- This is equivalent to **maximum likelihood estimation (MLE)**: we seek θ that maximises the probability of the observed corpus under the model
- **Connection to information theory**: minimising (2.1) is equivalent to minimising the KL divergence $D_{\text{KL}}(P_{\text{data}} \parallel \pi_{\theta})$. Indeed:

$$D_{\text{KL}}(P_{\text{data}} \parallel \pi_{\theta}) = \mathbb{E}_{P_{\text{data}}} [\log P_{\text{data}}(x_t \mid x_{<t}) - \log \pi_{\theta}(x_t \mid x_{<t})] = \underbrace{H(P_{\text{data}})}_{\text{const. in } \theta} + \mathcal{L}_{\text{PT}}(\theta)$$

Since $H(P_{\text{data}})$ does not depend on θ , $\arg \min_{\theta} D_{\text{KL}} = \arg \min_{\theta} \mathcal{L}_{\text{PT}}$

- **Teacher forcing and causal masking**: at each position t , the model is conditioned on the *true* preceding tokens $(x_1, \dots, x_{t-1})$, not on its own predictions. Because the ground-truth tokens are known at training time, the causal attention mask $M_{ij} = \mathbf{1}[j \leq i]$ allows the Transformer to compute $\pi_{\theta}(x_t \mid x_{<t})$ for all $t = 1, \dots, N$ in parallel, yielding $O(N)$ loss terms from a single $O(N^2 d)$ forward pass

Pre-training: Data and Scale

- Pre-training corpora are drawn from diverse web-scale sources:
 - ▶ **Common Crawl**: petabytes of raw web text (requires heavy filtering)
 - ▶ **Wikipedia, books, code repositories** (GitHub), scientific papers (arXiv)
 - ▶ Proprietary data for commercial models
- **Data quality pipeline**: raw text → language filtering → deduplication → quality scoring → toxicity filtering
- **Tokenisation**: recall from Week 1 that Byte Pair Encoding (BPE) converts raw text into subword tokens with $|\mathcal{V}| \approx 32,000\text{--}128,000$
- **Scaling laws** (Kaplan et al., 2020; Hoffmann et al., 2022): the pre-training loss decreases predictably as a power law in:
 - ▶ Model size (number of parameters)
 - ▶ Dataset size (number of tokens)
 - ▶ Compute budget (FLOPs)
- **Chinchilla scaling** (Hoffmann et al., 2022): for compute-optimal training, the number of tokens D should scale linearly with the number of parameters N , i.e. $D \propto N$

From Base Model to Assistant

- A pre-trained base model π_{base} is a **text completion engine**: given a prefix, it generates a plausible continuation
- **Problem**: base models do not naturally follow instructions
 - ▶ Input: “What is the capital of France?”
 - ▶ Base model output: “What is the capital of Germany? What is the capital of Spain? ...” (continues the pattern of questions)
- An **assistant model** should instead respond: “The capital of France is Paris.”
- The gap between base model behaviour and desired assistant behaviour motivates **fine-tuning**:
 - 1 **Supervised Fine-Tuning (SFT)**: teach the model the format and style of helpful responses using demonstration data
 - 2 **Reinforcement Learning from Human Feedback (RLHF)**: teach the model to distinguish good from bad responses using preference feedback
- The base model already *has* the knowledge (from pre-training); fine-tuning teaches it *when and how* to use that knowledge

The SFT Objective

- Let $\mathcal{D}_{\text{SFT}} = \{(x_q, y_q)\}_{q=1}^Q$ be a dataset of Q prompt-response pairs, where each prompt $x_q = (x_{q,1}, \dots, x_{q,S_q})$ is a token sequence of length S_q and each response $y_q = (y_{q,1}, \dots, y_{q,T_q})$ is a token sequence of length T_q
- The SFT loss is the conditional negative log-likelihood over *response* tokens only (with teacher forcing as in pre-training):

$$\mathcal{L}_{\text{SFT}}(\theta) = -\frac{1}{Q} \sum_{q=1}^Q \frac{1}{T_q} \sum_{t=1}^{T_q} \log \pi_{\theta}(y_{q,t} \mid x_q, y_{q,<t}) \quad (2.2)$$

where $y_{q,<t} = (y_{q,1}, \dots, y_{q,t-1})$ is the ground-truth prefix. The prompt tokens x_q appear in the conditioning but are *not* included in the loss ([loss masking](#))

- **Data quality:** SFT performance is highly sensitive to the quality of (x_q, y_q) pairs
 - ▶ **Diversity:** prompts should cover a wide range of tasks (QA, summarisation, coding, maths, etc.)
 - ▶ **Quality:** responses should be expert-written, accurate, and well-formatted
 - ▶ **Quantity:** a relatively small number of high-quality examples ($Q \sim 10^3\text{--}10^5$) can be effective (Zhou et al., 2023)

Low-Rank Structure of Fine-Tuning Updates

- Let $W_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ be a pre-trained weight matrix and W_{ft} the same matrix after full fine-tuning. Define the update $\Delta W = W_{\text{ft}} - W_0$
- The singular value decomposition (SVD) of ΔW is:

$$\Delta W = U \Sigma V^\top = \sum_{i=1}^{\min(d_{\text{out}}, d_{\text{in}})} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$$

where $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$ are the singular values

- Aghajanyan et al. (2021) observed that for fine-tuning on downstream tasks, the singular values σ_i decay rapidly. The **effective rank** — a measure of how spread out vs. peaked the distribution of singular values is —

$$r_{\text{eff}}(\Delta W) = \frac{(\sum_i \sigma_i)^2}{\sum_i \sigma_i^2} = \frac{\|\Delta W\|_*^2}{\|\Delta W\|_F^2}$$

satisfies $r_{\text{eff}} \ll \min(d_{\text{out}}, d_{\text{in}})$. For GPT-3 175B, $r_{\text{eff}} \leq 10$ for most weight matrices

- Because the singular values decay so fast, truncating the SVD to its top r components $\Delta W_r = U_r \Sigma_r V_r^\top$ retains most of the energy of ΔW (i.e. $\|\Delta W_r\|_F \approx \|\Delta W\|_F$). By the Eckart–Young theorem this is the best rank- r approximation in Frobenius norm, so a low-rank matrix can faithfully represent the fine-tuning update
- Note that $\Delta W_r = U_r \Sigma_r V_r^\top = (U_r \Sigma_r)(V_r^\top) = BA$ where $B \in \mathbb{R}^{d_{\text{out}} \times r}$ and $A \in \mathbb{R}^{r \times d_{\text{in}}}$. This motivates directly *parametrising* ΔW as a product of two low-rank factors BA and learning them during training $\Rightarrow$ **LoRA**

LoRA: Formulation

Definition 2.1 (Low-Rank Adaptation, LoRA (Hu et al., 2022))

Given a pre-trained weight matrix $W_0 \in \mathbb{R}^{d_{out} \times d_{in}}$ and input $x \in \mathbb{R}^{d_{in}}$, the adapted forward pass is:

$$h = W_0 x + \frac{\alpha}{r} B A x \quad (2.3)$$

where $B \in \mathbb{R}^{d_{out} \times r}$, $A \in \mathbb{R}^{r \times d_{in}}$, $r \ll \min(d_{out}, d_{in})$, and $\alpha > 0$ is a fixed scaling hyperparameter.

- W_0 is **frozen**; only (B, A) receive gradients. The effective update is $\Delta W = \frac{\alpha}{r} B A \in \mathbb{R}^{d_{out} \times d_{in}}$ with $\text{rank}(\Delta W) \leq r$
- **Trainable parameters per matrix**: $r(d_{out} + d_{in})$ instead of $d_{out} \cdot d_{in}$. The compression ratio is:

$$\frac{d_{out} \cdot d_{in}}{r(d_{out} + d_{in})} = \frac{d}{2r} \quad (\text{when } d_{out} = d_{in} = d)$$

For $d = 4096$, $r = 16$: ratio = 128×

LoRA: The α/r Scaling Factor

- **Problem:** at initialisation, $\|BA\|_F = \Theta(\sqrt{r})$ — doubling the rank increases the perturbation size, making hyperparameter tuning rank-dependent
- **Fix:** the $\frac{\alpha}{r}$ factor absorbs this rank-dependence, so the *same learning rate works across different values of r* without retuning
- **Two common choices of α :**
 - ▶ $\alpha = r$ (original LoRA default): update norm grows as $\sqrt{r}$. Higher rank = larger total update — useful when the task benefits from more capacity
 - ▶ $\alpha = \sqrt{r}$: update norm $\sim O(1)$. The total perturbation is rank-independent — better for controlled experiments that isolate the effect of rank from that of update magnitude

Why Not Just SFT?

- SFT teaches the model to *imitate* a fixed dataset of expert responses
- But imitation has a fundamental practical limit: **quality is expensive** — writing thousands of expert-quality responses requires significant human effort
- **A better approach**: instead of showing the model what a good answer looks like, *let the model try many answers and tell it which ones are better*
- This is the core idea of **reinforcement learning (RL)**: the model learns from *trial and error*, guided by a reward signal

Part 2: Reinforcement Learning Foundations

Goal: build the mathematical framework of reinforcement learning (RL) from scratch and specialise it to LLMs. **No prior RL knowledge is assumed.**

- What is reinforcement learning?
- Markov Decision Processes (MDPs)
- Policies, value functions, and the advantage function
- The policy gradient theorem and REINFORCE
- Generalised Advantage Estimation (GAE)
- Specialisation to LLMs: the KL-constrained objective

What is Reinforcement Learning? — The Idea

- Imagine training a dog. You cannot show the dog a “correct walk” to imitate (that would be supervised learning). Instead, you let the dog try different behaviours and give it a treat when it does something good. Over time, the dog learns which behaviours lead to treats
- In the LLM setting: the model generates a response (a sequence of actions), and then receives a **score** (reward) indicating how good the response was. Over many trials, it learns to generate higher-scoring responses
- The key elements:
 - An **agent** (the model π_θ) takes **actions** (generates tokens) in an **environment**
 - After completing a sequence of actions, the agent receives a scalar **reward** $R \in \mathbb{R}$
 - The goal is to find parameters θ that maximise the **expected cumulative reward**:

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=1}^T r_t \right]$$

where $\tau = (s_1, a_1, r_1, \dots, s_T, a_T, r_T)$ is a **trajectory** (a full episode of interaction)

- Unlike SFT, where the loss compares the model's output directly to a target y_q , in RL there is *no target* — just a scalar reward that says “how well did you do overall?” The model must *explore* different actions to discover which ones lead to high reward

Markov Decision Processes

Definition 2.2 (Markov Decision Process (MDP))

An MDP is a tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma, T)$ where:

- $\mathcal{S}$ is the **state space**
- $\mathcal{A}$ is the **action space**
- $P(s' | s, a)$ is the **transition kernel**: probability of moving to state s' given state s and action a
- $R(s, a) \in \mathbb{R}$ is the **reward function**
- $\gamma \in [0, 1]$ is the **discount factor**
- T is the **horizon** (episode length)

A **policy** $\pi(a | s)$ is a conditional distribution over actions given states. The agent's goal is to find a policy π^* that maximises $J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=1}^T \gamma^{t-1} r_t \right]$.

- **In plain English**: the agent is in some situation (state), picks an action, receives a reward, and moves to a new situation. The policy is its decision-making rule. The discount factor γ controls how much the agent cares about future vs. immediate rewards ($\gamma = 1$: equal weight; $\gamma \rightarrow 0$: myopic)
- The **Markov property**: $P(s_{t+1} | s_1, a_1, \dots, s_t, a_t) = P(s_{t+1} | s_t, a_t)$ — the future depends on the present state and action only, not on the full history

LLM Text Generation as an MDP

- Let us specialise the MDP framework to autoregressive text generation:
 - ▶ **State** at time t : $s_t = (x, y_1, \dots, y_{t-1}) \in \mathcal{V}^*$ (prompt x concatenated with tokens generated so far)
 - ▶ **Action** at time t : $a_t = y_t \in \mathcal{V}$ (next token chosen from the vocabulary)
 - ▶ **Policy**: $\pi_\theta(a_t | s_t) = \pi_\theta(y_t | x, y_{<t})$ (the language model's conditional distribution)
 - ▶ **Transition**: deterministic concatenation; $s_{t+1} = (s_t, a_t) = (x, y_1, \dots, y_t)$
 - ▶ **Reward**: typically sparse and terminal; $r_t = 0$ for $t < T$ and $r_T = R(x, y)$ where R is a reward model scoring the complete response $y = (y_1, \dots, y_T)$
 - ▶ **Discount**: $\gamma = 1$ (undiscounted, since episodes are finite)
- The horizon T is the response length; the episode terminates when $y_T = \text{<eos>}$
- Notice that the transition is deterministic and the state grows by one token per step — all stochasticity comes from the policy π_θ itself. This is a much simpler MDP than typical RL environments (robotics, games)

Reward Hacking and the Need for Regularisation

- We have framed LLM generation as an MDP, and the natural objective is to maximise expected reward. But a naive approach fails
- Consider the unconstrained objective $\max_{\theta} \mathbb{E}_{x \sim \mathcal{D}} \mathbb{E}_{y \sim \pi_{\theta}(\cdot|x)}[R(x, y)]$
- Since R is a learned approximation $R_{\psi} \approx R^*$, the policy will exploit errors in R_{ψ} — concentrating mass on outputs where R_{ψ} overestimates R^* . This is **reward hacking**
- **Solution**: constrain the policy to remain close to a reference π_{ref} (typically π_{SFT}), so that π_{θ} cannot move into regions where R_{ψ} is unreliable

The KL-Constrained RL Objective

- We want to maximise reward but *not stray too far* from the SFT model π_{ref} . The KL divergence $D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}})$ measures how different the current policy is from the reference, so we add it as a penalty
- The **KL-regularised objective** adds a divergence penalty to the reward:

$$\max_{\theta} \mathbb{E}_{x \sim \mathcal{D}} \mathbb{E}_{y \sim \pi_{\theta}(\cdot|x)} \left[R(x, y) - \beta D_{\text{KL}}(\pi_{\theta}(\cdot|x) \parallel \pi_{\text{ref}}(\cdot|x)) \right] \quad (2.4)$$

- The coefficient $\beta > 0$ controls the regularisation strength: large β keeps $\pi_{\theta} \approx \pi_{\text{ref}}$; small β allows larger deviations
- **Per-token decomposition**: both π_{θ} and π_{ref} factorise autoregressively:
 $\pi(y|x) = \prod_{t=1}^T \pi(y_t|s_t)$. Therefore $\log \frac{\pi_{\theta}(y|x)}{\pi_{\text{ref}}(y|x)} = \sum_{t=1}^T \log \frac{\pi_{\theta}(y_t|s_t)}{\pi_{\text{ref}}(y_t|s_t)}$, and taking expectations:

$$D_{\text{KL}}(\pi_{\theta}(\cdot|x) \parallel \pi_{\text{ref}}(\cdot|x)) = \mathbb{E}_{y \sim \pi_{\theta}} \left[\log \frac{\pi_{\theta}(y|x)}{\pi_{\text{ref}}(y|x)} \right] = \mathbb{E}_{y \sim \pi_{\theta}} \left[\sum_{t=1}^T \log \frac{\pi_{\theta}(y_t|s_t)}{\pi_{\text{ref}}(y_t|s_t)} \right]$$

- This per-token form defines an **effective per-token reward**:

$$\tilde{r}_t = -\beta \log \frac{\pi_{\theta}(y_t|s_t)}{\pi_{\text{ref}}(y_t|s_t)}, \quad t < T; \quad \tilde{r}_T = R(x, y) - \beta \log \frac{\pi_{\theta}(y_T|s_T)}{\pi_{\text{ref}}(y_T|s_T)}$$

The original objective is now $\max_{\theta} \mathbb{E}[\sum_{t=1}^T \tilde{r}_t]$, which has the form of maximising a cumulative return in a standard MDP. This means we can directly apply off-the-shelf RL algorithms (e.g. PPO) using $\tilde{r}_t$ as the per-step reward

The Optimisation Challenge

- We want to maximise the expected return:

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=1}^T r_t \right] = \sum_{\tau} p_\theta(\tau) R(\tau)$$

where $\tau = (s_1, a_1, r_1, \dots, s_T, a_T, r_T)$ is a trajectory sampled by rolling out π_θ , and $R(\tau) = \sum_{t=1}^T r_t$ is its total reward

- The trajectory probability is $p_\theta(\tau) = \prod_{t=1}^T \pi_\theta(a_t | s_t) \cdot P(s_{t+1} | s_t, a_t)$. In the LLM setting, the transitions are deterministic ($s_{t+1} = (s_t, a_t)$), so $p_\theta(\tau) = \prod_{t=1}^T \pi_\theta(a_t | s_t)$
- **Problem:** $J(\theta)$ is an expectation over *discrete* sequences $\tau \in \mathcal{V}^T$. We cannot compute $\nabla_\theta J(\theta)$ by backpropagating through the sampling operation (sampling is not differentiable)
- Recall: in SFT, the model outputs a raw score vector $z_t \in \mathbb{R}^{|\mathcal{V}|}$ and we compute $\pi_\theta(y_t | s_t) = \text{softmax}(z_t)_{y_t}$. The loss $\mathcal{L} = -\sum_t \log \pi_\theta(y_t^* | s_t)$ is evaluated at *fixed* target tokens y_t^* — the chain $\theta \rightarrow z_t \rightarrow \text{softmax} \rightarrow \mathcal{L}$ is fully differentiable, so $\nabla_\theta \mathcal{L}$ is obtained by standard backpropagation. In RL, the reward depends on *sampled* tokens $y_t \sim \text{softmax}(z_t)$, and sampling is a discrete, non-differentiable operation — we cannot compute $\partial y_t / \partial \theta$
- The **policy gradient theorem** (Williams, 1992) gets around this: it computes $\nabla_\theta J(\theta)$ from sampled trajectories without differentiating through the sampling step

The Log-Derivative Trick

- The key identity is the **log-derivative trick** (also called the score function estimator). For any differentiable $p_\theta(\tau) > 0$:

$$\nabla_\theta p_\theta(\tau) = p_\theta(\tau) \cdot \nabla_\theta \log p_\theta(\tau)$$

This follows from $\nabla_\theta \log p_\theta(\tau) = \frac{\nabla_\theta p_\theta(\tau)}{p_\theta(\tau)}$

- Applying this to $J(\theta)$:

$$\begin{aligned}\nabla_\theta J(\theta) &= \sum_{\tau} \nabla_\theta p_\theta(\tau) R(\tau) = \sum_{\tau} p_\theta(\tau) \nabla_\theta \log p_\theta(\tau) R(\tau) \\ &= \mathbb{E}_{\tau \sim p_\theta} [\nabla_\theta \log p_\theta(\tau) R(\tau)]\end{aligned}\tag{2.5}$$

- The gradient is now an *expectation* under p_θ — we can estimate it by sampling trajectories from π_θ
- So to compute the gradient: (1) generate several responses from the model, (2) score each one, (3) for each response, compute $\nabla_\theta \log p_\theta(\tau)$ (which *is* differentiable — it just involves the model's log-probabilities), and (4) weight it by the reward. We never differentiate through the sampling step itself

The Policy Gradient Theorem (1/2)

- Substituting $\log p_\theta(\tau) = \sum_{t=1}^T \log \pi_\theta(a_t | s_t)$ into (2.5):

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim p_\theta} \left[\left(\sum_{t=1}^T \nabla_\theta \log \pi_\theta(a_t | s_t) \right) R(\tau) \right]$$

- Causality argument:** the term $\nabla_\theta \log \pi_\theta(a_t | s_t)$ at time t is multiplied by the *full* return $R(\tau) = \sum_{t'=1}^T r_{t'}$. But for $t' < t$, the reward $r_{t'}$ has already been determined before a_t is chosen, so $r_{t'}$ is a constant with respect to the expectation over a_t . We can therefore pull it out:

$$\begin{aligned} \mathbb{E}_{a_t \sim \pi_\theta(\cdot | s_t)} [\nabla_\theta \log \pi_\theta(a_t | s_t) \cdot r_{t'}] &= r_{t'} \cdot \mathbb{E}_{a_t} [\nabla_\theta \log \pi_\theta(a_t | s_t)] \\ &= r_{t'} \cdot \sum_{a_t} \pi_\theta(a_t | s_t) \cdot \frac{\nabla_\theta \pi_\theta(a_t | s_t)}{\pi_\theta(a_t | s_t)} \\ &= r_{t'} \cdot \underbrace{\nabla_\theta \sum_{a_t} \pi_\theta(a_t | s_t)}_{=1} = 0 \end{aligned}$$

The Policy Gradient Theorem (2/2)

- Dropping these zero-expectation terms yields the **REINFORCE** gradient:

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\pi_{\theta}} \left[\sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) G_t \right] \quad (2.6)$$

where $G_t = \sum_{t'=t}^T r_{t'}$ is the **return-to-go** from step t

- The intuition is simple: increase the log-probability of action a_t in proportion to how much future reward G_t followed
 - ▶ If a token was followed by high total reward $\Rightarrow$ make that token *more likely* next time
 - ▶ If a token was followed by low total reward $\Rightarrow$ make that token *less likely* next time
 - ▶ This is trial-and-error learning: actions that led to good outcomes are **reinforced**

The REINFORCE Algorithm

- **REINFORCE** (Williams, 1992) is the simplest policy gradient algorithm. It estimates (2.6) via Monte Carlo sampling:
 - ① Sample a trajectory $\tau = (s_1, a_1, r_1, \dots, s_T, a_T, r_T)$ by rolling out π_θ
 - ② Compute the return-to-go $G_t = \sum_{t'=t}^T r_{t'}$ for each $t = 1, \dots, T$
 - ③ Compute the gradient estimate: $\hat{g} = \sum_{t=1}^T \nabla_\theta \log \pi_\theta(a_t | s_t) G_t$
 - ④ Update parameters: $\theta \leftarrow \theta + \alpha \hat{g}$
- **Unbiasedness**: $\mathbb{E}[\hat{g}] = \nabla_\theta J(\theta)$ by construction from (2.6)
- **High variance**: each term in $\hat{g}$ is a product $\nabla_\theta \log \pi_\theta(a_t | s_t) \cdot G_t$. Two independent sources of randomness compound:
 - ▶ $G_t = \sum_{t'=t}^T r_{t'}$ sums rewards over all future tokens — each sampled from a vocabulary of size $|\mathcal{V}| \sim 10^5$, so G_t can vary wildly between trajectories
 - ▶ $\nabla_\theta \log \pi_\theta(a_t | s_t)$ depends on which token a_t was sampled; different tokens give gradient vectors pointing in very different directions

Since $\text{Var}(XY) = \text{Var}(X)\text{Var}(Y) + \dots$ for independent variables, the product amplifies both sources. Summing $T \sim 10^2\text{--}10^3$ such terms makes the total variance prohibitively large

- If the gradient estimate fluctuates wildly from sample to sample, the parameter updates “jump around” rather than moving steadily toward a good policy. Training becomes slow and unstable
- This motivates (i) variance reduction via baselines and advantage estimation, and (ii) constrained updates via PPO (Part 3)

Variance Reduction with Baselines

- REINFORCE weights each token's gradient by the total future reward G_t . But G_t can be large even for “average” actions — what matters is whether G_t is *above or below* what we would typically expect from state s_t . Subtracting a **baseline** (“what we normally get from this state”) centres the signal and reduces noise
- Formally: we subtract $b(s_t)$ from the return:

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\pi_{\theta}} \left[\sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) (G_t - b(s_t)) \right] \quad (2.7)$$

- **Unbiasedness**: for any $b(s_t)$ depending only on s_t (not on a_t), the subtraction does not introduce bias. Proof:

$$\mathbb{E}_{a_t \sim \pi_{\theta}(\cdot | s_t)} [\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) b(s_t)] = b(s_t) \underbrace{\nabla_{\theta} \sum_{a_t} \pi_{\theta}(a_t | s_t)}_{=1} = 0$$

- **Optimal baseline**: write $g_t = \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$. Minimising the second moment $\mathbb{E}_{a_t} [\|g_t\|^2 (G_t - b)^2]$ w.r.t. b :

$$\frac{\partial}{\partial b} \mathbb{E}_{a_t} [\|g_t\|^2 (G_t - b)^2] = -2 \mathbb{E}_{a_t} [\|g_t\|^2 (G_t - b)] \stackrel{!}{=} 0 \implies b^*(s_t) = \frac{\mathbb{E}_{a_t} [\|g_t\|^2 G_t]}{\mathbb{E}_{a_t} [\|g_t\|^2]}$$

When $\|g_t\|^2$ is approximately constant across actions, $b^*(s_t) \approx \mathbb{E}_{\pi} [G_t | s_t] = V^{\pi}(s_t)$

Value Functions and the Advantage

Definition 2.3 (Value, Action-Value, and Advantage Functions)

For a policy π :

- **State-value:** $V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t'=t}^T r_{t'} \mid s_t = s \right]$ (“how good is this state on average?”)
- **Action-value:** $Q^\pi(s, a) = \mathbb{E}_\pi \left[\sum_{t'=t}^T r_{t'} \mid s_t = s, a_t = a \right]$ (“how good is taking action a in state s ?”)
- **Advantage:** $A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$ (“how much better is action a compared to the average action in state s ?”)

- **Relationships:** by definition $V^\pi(s) = \mathbb{E}_{a \sim \pi(\cdot|s)}[Q^\pi(s, a)]$, so $\mathbb{E}_{a \sim \pi(\cdot|s)}[A^\pi(s, a)] = \mathbb{E}_{a \sim \pi}[Q^\pi(s, a)] - V^\pi(s) = V^\pi(s) - V^\pi(s) = 0$
- **Advantage form of the policy gradient:** substituting $b(s_t) = V^\pi(s_t)$ into (2.7) gives:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=1}^T \nabla_\theta \log \pi_\theta(a_t \mid s_t) A^{\pi_\theta}(s_t, a_t) \right]$$

since $\mathbb{E}[G_t \mid s_t, a_t] = Q^\pi(s_t, a_t)$ by definition of Q^π , so $\mathbb{E}[G_t - V^\pi(s_t) \mid s_t, a_t] = A^\pi(s_t, a_t)$

- The gradient is now weighted by A^π : tokens with $A^\pi > 0$ (better than average under π) have their probability increased; tokens with $A^\pi < 0$ have their probability decreased. Tokens with $A^\pi \approx 0$ contribute negligible gradient — this is the variance reduction mechanism

Estimating the Advantage: Temporal Difference Residual

- Computing $A^\pi(s_t, a_t)$ requires knowing $V^\pi(s_t)$ — the expected total reward from state s_t — but this is unknown
- We learn a parametric approximation $V_\phi(s) \approx V^\pi(s)$ (the **critic**, trained by regression on observed returns)
- The **temporal difference (TD) residual** provides a one-step estimate of the advantage:

$$\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t) \quad (2.8)$$

where $\gamma \in [0, 1]$ is the discount factor ($\gamma = 1$ in the undiscounted LLM setting)

- **Why δ_t estimates A^π :** recall $Q^\pi(s_t, a_t) = \mathbb{E}_\pi[r_t + \gamma V^\pi(s_{t+1}) \mid s_t, a_t]$. After taking action a_t and observing r_t, s_{t+1} , we have the one-sample estimate $Q^\pi(s_t, a_t) \approx r_t + \gamma V^\pi(s_{t+1})$. Since $A^\pi = Q^\pi - V^\pi$:

$$A^\pi(s_t, a_t) \approx r_t + \gamma V^\pi(s_{t+1}) - V^\pi(s_t)$$

Replacing the unknown V^π with the learned V_ϕ gives exactly δ_t

- **Bias–variance trade-off:**
 - ▶ **Low variance:** δ_t uses only one step of actual reward r_t , then bootstraps from V_ϕ — no sum over future randomness
 - ▶ **Biased:** $\delta_t = A^\pi(s_t, a_t) + \gamma(V_\phi(s_{t+1}) - V^\pi(s_{t+1})) - (V_\phi(s_t) - V^\pi(s_t))$, so the error is proportional to $\|V_\phi - V^\pi\|$. If the critic is inaccurate, δ_t is a poor estimate

Generalised Advantage Estimation (GAE)

- The TD residual δ_t uses one step of real reward then bootstraps from V_ϕ — low variance (one random term) but biased (relies on $V_\phi \approx V^\pi$). Using more real steps reduces bias (less reliance on V_ϕ) but increases variance (more random terms in the sum)
- **GAE** (Schulman et al., 2016) blends all horizons via an exponentially-weighted sum of TD residuals, controlled by $\lambda \in [0, 1]$:

$$\hat{A}_t^{\text{GAE}(\gamma, \lambda)} = \sum_{\ell=0}^{T-t} (\gamma \lambda)^\ell \delta_{t+\ell} \quad (2.9)$$

- λ controls the **bias–variance trade-off**:
 - ▶ $\lambda = 0$: $\hat{A}_t = \delta_t$ — one-step, low variance, high bias
 - ▶ $\lambda = 1$: $\hat{A}_t = G_t - V_\phi(s_t)$ — full return, high variance, low bias
- **LLM-RLHF**: $\lambda \in [0.95, 0.99]$, $\gamma = 1$. High λ is preferred because the critic V_ϕ may be inaccurate early in training, so low bias matters more than low variance

Putting It Together: The LLM RL Objective

- Combining the KL-regularised per-token rewards $\tilde{r}_t$ with GAE, the policy gradient for LLM-RLHF is:

$$\nabla_{\theta} J(\theta) \approx \mathbb{E}_{\mathbf{x}, y \sim \pi_{\theta}} \left[\sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(y_t | s_t) \hat{A}_t^{\text{GAE}} \right]$$

where $\hat{A}_t^{\text{GAE}} = \sum_{\ell=0}^{T-t} \lambda^{\ell} \tilde{\delta}_{t+\ell}$ and $\tilde{\delta}_t = \tilde{r}_t + V_{\phi}(s_{t+1}) - V_{\phi}(s_t)$

- Remaining problem:** nothing constrains the size of the update. If one gradient step makes π_{θ} assign very different probabilities to tokens than before, the model can start producing degenerate text and never recover
- PPO** prevents this by bounding how much $\pi_{\theta}(y_t | s_t)$ can change relative to the previous policy in a single update
- GRPO** removes the need to train a separate critic V_{ϕ} by estimating advantages from groups of sampled responses

Part 3: RLHF — PPO, GRPO, and the Training Pipeline

- **PPO** (Schulman et al., 2017) is the principal algorithm used for RLHF in models such as ChatGPT (Ouyang et al., 2022)
- **Why constrain updates?** Unlike supervised learning, in RL the policy π_θ determines *which data* is collected. A bad update $\Rightarrow$ poor trajectories $\Rightarrow$ biased gradient estimates $\Rightarrow$ worse updates: a **vicious cycle** leading to policy collapse
- **Core idea**: constrain each update so that π_θ cannot change too much at once, by clipping the probability ratio between the new and old policies
- Before presenting PPO's clipping mechanism, we first examine its predecessor **TRPO** — the theoretically principled (but expensive) approach to the same problem

Trust Region Policy Optimisation (TRPO)

- How do we formalise “don’t change the policy too much”? **TRPO** (Schulman et al., 2015) defines a *trust region* — a KL ball around the current policy within which the surrogate objective reliably approximates true performance:

$$\begin{aligned} \max_{\theta} \quad & L^{\text{CPI}}(\theta) := \mathbb{E}_{\pi_{\theta_{\text{old}}}} \left[\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)} \hat{A}_t \right] \\ \text{s.t.} \quad & \mathbb{E}_{s \sim \rho_{\theta_{\text{old}}}} [D_{\text{KL}}(\pi_{\theta_{\text{old}}}(\cdot | s) \parallel \pi_{\theta}(\cdot | s))] \leq \delta \end{aligned} \quad (2.10)$$

where L^{CPI} is the *surrogate objective* (“conservative policy iteration”) and $\delta > 0$ is the trust region radius

- The family $\{\pi_{\theta}(\cdot | s) : \theta \in \Theta\}$ is a **statistical manifold**. The KL divergence induces a Riemannian metric on Θ : to second order,

$$D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\theta+d\theta}) = \frac{1}{2} d\theta^{\top} F(\theta) d\theta + O(\|d\theta\|^3)$$

where $F(\theta) = \mathbb{E}_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta} (\nabla_{\theta} \log \pi_{\theta})^{\top}]$ is the **Fisher information matrix**

- Hence F is the metric tensor of the manifold: it measures how fast the distribution changes in each parameter direction

TRPO: Natural Gradient

- The steepest-ascent direction of $L^{\text{CPI}}(\theta)$ under the Fisher–Rao metric (i.e. per unit of KL divergence) is the **natural gradient**:

$$\tilde{\nabla}_{\theta} L^{\text{CPI}} = F(\theta)^{-1} \nabla_{\theta} L^{\text{CPI}}(\theta)$$

This follows from the general result: the steepest-ascent direction of f w.r.t. a metric G is $G^{-1} \nabla f$ (cf. Riemannian gradient on (M, g))

- TRPO solves (2.10) approximately via:
 - ① Policy gradient $g = \nabla_{\theta} L^{\text{CPI}}(\theta)$
 - ② Natural gradient step: $\Delta\theta = F^{-1}g$
 - ③ The quadratic approximation $D_{\text{KL}} \approx \frac{1}{2} \Delta\theta^{\top} F \Delta\theta$ gives the largest step along $F^{-1}g$ that satisfies $D_{\text{KL}} \leq \delta$. Setting $\frac{1}{2} \Delta\theta^{\top} F \Delta\theta = \delta$ and solving:
$$\Delta\theta \leftarrow \sqrt{2\delta / (\Delta\theta^{\top} F \Delta\theta)} \Delta\theta$$

TRPO: Practical Limitations

- TRPO provides strong theoretical guarantees (monotonic policy improvement under certain conditions)
- However, it has significant **practical limitations**:
 - ▶ **Computational cost**: the Fisher information matrix F is $d \times d$ where d is the number of parameters. For a 7B model, $d \approx 7 \times 10^9$ — F cannot be stored, let alone inverted
 - ▶ **Conjugate gradient**: since forming F is infeasible, TRPO computes $F^{-1}g$ via the *conjugate gradient* (CG) algorithm — an iterative method that solves $Fx = g$ using only matrix-vector products Fv (computed efficiently via automatic differentiation), without ever forming F . Still expensive: requires ~ 10 such products per update step
- Can we achieve similar stability guarantees with a *first-order* method? Yes — this is exactly what PPO's clipping mechanism provides

From TRPO to PPO: The Clipping Idea

- Define the **importance sampling ratio** $\rho_t(\theta) = \pi_\theta(a_t|s_t) / \pi_{\theta_{\text{old}}}(a_t|s_t)$
- Instead of TRPO's KL constraint (expensive, second-order), PPO simply *clips* ρ_t to $[1 - \varepsilon, 1 + \varepsilon]$:

$$L^{\text{CLIP}}(\theta) = \mathbb{E} \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_t \right) \right] \quad (2.11)$$

where $\varepsilon \in [0.1, 0.2]$ is a hyperparameter

- **Interpretation:** once the policy moves too far from $\pi_{\theta_{\text{old}}}$ (i.e. ρ_t leaves $[1 - \varepsilon, 1 + \varepsilon]$), the gradient is zeroed out — no further incentive to move
- $L^{\text{CLIP}}(\theta) \leq L^{\text{CPI}}(\theta)$ with equality at $\theta = \theta_{\text{old}}$, so maximising L^{CLIP} amounts to maximising a *pessimistic lower bound* on the true surrogate — the same conservative update guarantee as TRPO
- **Advantages over TRPO:** first-order only (standard SGD), trivial to implement, compatible with minibatch training and multiple epochs per batch

PPO Training Loop for LLMs

- The full PPO training procedure for RLHF:

- 1 **Collect trajectories**: sample a batch of prompts $\{x_i\}$ from $\mathcal{D}$; for each prompt, generate response $y_i \sim \pi_{\theta_{\text{old}}}(\cdot|x_i)$
- 2 **Score**: compute reward $R(x_i, y_i)$ from the reward model and per-token KL penalties
- 3 **Compute advantages**: use GAE (2.9) with the critic network V_ϕ :

$$\hat{A}_t = \sum_{\ell=0}^{T-t} (\gamma\lambda)^\ell \delta_{t+\ell}, \quad \delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$$

- 4 **Optimise**: for K epochs of minibatch SGD, update:
 - ★ Policy θ by maximising $L^{\text{CLIP}}(\theta)$
 - ★ Critic ϕ by minimising $\|V_\phi(s_t) - \hat{V}_t^{\text{target}}\|^2$
 - 5 Set $\theta_{\text{old}} \leftarrow \theta$ and return to step 1
- Typical hyperparameters: $K = 4$ epochs, $\varepsilon = 0.2$, $\lambda = 0.95$, $\gamma = 1.0$
 - The critic V_ϕ is typically initialised from the reward model or a copy of the policy, and outputs a scalar value estimate per token position
 - **Drawback**: the critic *doubles* the memory and compute cost — for a 70B-parameter LLM, one must also maintain a 70B-parameter critic. **This motivates critic-free methods** such as GRPO

GRPO: The Key Idea

- **Problem:** PPO needs a critic V_ϕ to compute advantages $\hat{A}_t = Q(s_t, a_t) - V_\phi(s_t)$. This doubles memory
- **GRPO's insight** (Guo et al., 2025): instead of *learning* a baseline V_ϕ , *estimate* it by sampling. For each prompt x_q , generate a *group* of G responses:

$$\{y_{q,1}, \dots, y_{q,G}\} \sim \pi_{\theta_{\text{old}}}(\cdot \mid x_q)$$

and score each one: $R(x_q, y_{q,g})$ for $g = 1, \dots, G$

- The advantage $A^\pi(x, y) = Q^\pi(x, y) - V^\pi(x)$ is then estimated as:
 - ▶ $Q^\pi(x_q, y_{q,g}) \approx R(x_q, y_{q,g})$ (observed reward)
 - ▶ $V^\pi(x_q) \approx \bar{R}_q = \frac{1}{G} \sum_{g'=1}^G R(x_q, y_{q,g'})$ (group mean as Monte Carlo baseline)
- This is unbiased: by the law of large numbers, $\bar{R}_q \rightarrow V^{\pi_{\theta_{\text{old}}}}(x_q)$ as $G \rightarrow \infty$
- **No critic network is needed:** the advantage is computed purely from group statistics

GRPO: Advantage and Loss

- The **group-normalised advantage** for response g is:

$$\hat{A}(x_q, y_{q,g}) = \frac{R(x_q, y_{q,g}) - \bar{R}_q}{\sigma_{R_q}} \quad (2.12)$$

where $\sigma_{R_q} = \text{std}(\{R(x_q, y_{q,g'})\}_{g'})$. Division by σ_{R_q} gives unit variance across the group, stabilising gradient magnitudes

- $\hat{A} > 0$ for responses scoring above the group average; $\hat{A} < 0$ for those below. The policy is pushed to produce more of the former and less of the latter
- The GRPO loss averages over prompts q , responses g in each group, and tokens t in each response:

$$\mathcal{L}_{\text{GRPO}}(\theta) = \mathbb{E}_q \frac{1}{G} \sum_{g=1}^G \frac{1}{T_g} \sum_{t=1}^{T_g} \left[\underbrace{\min(\rho_{t,g} \hat{A}_{q,g}, \text{clip}(\rho_{t,g}, 1-\varepsilon, 1+\varepsilon) \hat{A}_{q,g})}_{\text{PPO-Clip applied per token}} - \underbrace{\beta D_{\text{KL}}^{(t)}(\pi_\theta \| \pi_{\text{ref}})}_{\text{KL penalty from } \pi_{\text{ref}}} \right] \quad (2.13)$$

- Reading the formula:
 - $\rho_{t,g} = \pi_\theta(y_t|s_t)/\pi_{\theta_{\text{old}}}(y_t|s_t)$: per-token importance ratio (same as PPO)
 - $\hat{A}_{q,g}$: group-normalised advantage — *constant across all tokens* in response g (unlike PPO, where each token gets a different advantage)
 - The min/clip term is exactly PPO-Clip: it caps how much $\rho_{t,g}$ can deviate from 1
 - $\beta D_{\text{KL}}^{(t)}$: per-token KL divergence from the reference policy, preventing drift

Reinforcement Learning from Human Feedback (RLHF)

- **RLHF** (Ouyang et al., 2022) puts the pieces together — it is the end-to-end pipeline that turns π_{SFT} into an aligned assistant using the RL machinery from this part:
 - 1 **Train a reward model** R_ψ : collect human preference pairs ($y_w \succ y_l$) and fit R_ψ using the **Bradley–Terry model** (Bradley & Terry, 1952) — a pairwise comparison model where the probability that y_w is preferred over y_l is $P(y_w \succ y_l) = \sigma(R_\psi(x, y_w) - R_\psi(x, y_l))$, trained via binary cross-entropy
 - 2 **Optimise the policy**: use R_ψ as the reward in the KL-constrained objective (2.4), and update π_θ via **PPO** (clipped surrogate + critic) or **GRPO** (group-normalised advantages, no critic)

Part 4: Knowledge Distillation

- GRPO produced DeepSeek-R1 (671B MoE) — a powerful reasoner, but expensive to serve. How do we make these capabilities *accessible*?
- Knowledge distillation** (Hinton et al., 2015) trains a small *student* π_S to mimic a large *teacher* π_T
- Let $\mathbf{z}^T, \mathbf{z}^S \in \mathbb{R}^V$ be the teacher's and student's **logits** (pre-softmax outputs over the vocabulary) for a given input. The **softened distribution** at temperature $\tau > 0$ is:

$$p_i^{(\tau)} = \frac{\exp(z_i/\tau)}{\sum_{j=1}^V \exp(z_j/\tau)}$$

At $\tau = 1$ this is the standard softmax. As τ increases, the distribution flattens: the teacher's full ranking over all V tokens — not just the identity of the top token — is revealed. Hinton calls this the teacher's “dark knowledge”

- The **distillation loss** combines two objectives:

$$\mathcal{L}_{\text{KD}} = \underbrace{(1 - \alpha) \mathcal{L}_{\text{CE}}(y, p_S^{(1)})}_{\text{hard target: match ground truth } y} + \underbrace{\alpha \tau^2 D_{\text{KL}}(p_T^{(\tau)} \| p_S^{(\tau)})}_{\text{soft target: match teacher's distribution}}$$

where $\alpha \in [0, 1]$ balances the two terms. The τ^2 factor is needed because $\nabla_{z_i^S} D_{\text{KL}}$ scales as $1/\tau^2$ when the softened gradients shrink as the distribution flattens; the rescaling restores gradient magnitudes to a level comparable with the hard-target term

DeepSeek-R1: Distillation at Scale

- The general KD framework requires access to the teacher's logits $\mathbf{z}^T$. DeepSeek-R1 (671B MoE) uses a simpler approach: **distillation via SFT on chain-of-thought traces**
- **Data generation**: R1 generates candidate responses for each prompt, including its full reasoning chain. Responses are filtered by correctness (**rejection sampling**): only those arriving at the right answer are kept. This yields $\sim 800\text{K}$ samples ($\sim 600\text{K}$ reasoning tasks + $\sim 200\text{K}$ general instructions)
- **Student training**: open-source base models (Qwen2.5, Llama-3) are fine-tuned on these samples for 2 epochs using the standard SFT loss (2.2). **No logit matching, no KL divergence, no RL**

Student model	Base architecture	AIME 2024	MATH-500
R1-Distill-Qwen-7B	Qwen2.5-Math-7B	55.5	92.8
R1-Distill-Qwen-14B	Qwen2.5-14B	69.7	93.9
R1-Distill-Qwen-32B	Qwen2.5-32B	72.6	94.3
R1-Distill-Llama-70B	Llama-3.3-70B	70.0	94.5
GPT-4o	—	9.3	—
OpenAI o1-mini	—	63.6	90.0

- The **14B student surpasses o1-mini** on AIME; the 7B model outperforms GPT-4o by $6\times$
- **Key insight**: the reasoning traces carry the distillation signal — the student learns *how to think step-by-step*, not just *what to answer*. The reasoning process (not the logits) is the “dark knowledge”

The DeepSeek Shock: Market Impact

- On **27 January 2025**, DeepSeek-R1 was released as open-source. The base model (DeepSeek-V3) was reportedly trained on $\sim 2,048$ NVIDIA H800 GPUs at a compute cost of $\sim \$5.6\text{M}$
- **NVIDIA (NVDA) fell $\sim 17\%$ in a single session**, erasing $\sim \$589$ billion in market capitalisation — the **largest single-day loss for any company in US stock market history**
- NVIDIA's market cap dropped from $\sim \$3.5\text{T}$ to $\sim \$2.9\text{T}$. The Nasdaq fell 3.1%; semiconductor and data-centre stocks (Dell, Oracle, Super Micro) dropped 8–12%
- **The narrative**: DeepSeek challenged the “insatiable GPU demand” thesis underpinning NVIDIA's valuation. If frontier AI can be built with far fewer chips, the hardware capex implied by the scaling paradigm may not be as large as priced in
- **The rebound**: NVDA recovered $\sim 8\%$ the following day. The longer-term question: does efficiency *reduce* total compute demand, or *increase* it via Jevons' paradox?

Distilling Transformers into SSMs

- Recall from Week 1: **SSMs** (e.g. Mamba) offer $O(T)$ inference and fixed memory, but lag behind Transformers on benchmarks. **Can we distill a Transformer teacher into an SSM student?**
- The key observation:** both architectures define **sequence mixing matrices**. Attention produces $Y = M_{\text{attn}} V$ where $(M_{\text{attn}})_{ij} = \text{softmax}(QK^\top)_{ij}$. An SSM produces $Y = M_{\text{ssm}} X$ where $(M_{\text{ssm}})_{ij} = C_i \bar{A}^{i-j} \bar{B}_j$ for $j \leq i$ (and 0 otherwise). Distillation aligns $M_{\text{ssm}} \approx M_{\text{attn}}$
- MOHAWK** (Bick et al., NeurIPS 2024): progressive distillation in three stages:
 - Matrix orientation:** fit the SSM parameters $(\bar{A}, \bar{B}, C)$ so that M_{ssm} approximates the teacher's M_{attn} , layer by layer
 - Hidden-state alignment:** train each student block to reproduce the teacher's hidden representations, propagating real teacher activations as input
 - End-to-end KD:** fine-tune the full student with $\mathcal{L}_{\text{KD}}$ on the output distribution

Result: distilled Phi-1.5 (1.3B Transformer) into **Phi-Mamba** using only 3B tokens ($< 1\%$ of pre-training data)

- Llamba** (Bick et al., 2025): replaces all attention layers in Llama-3.1-8B-Instruct with Mamba-2 layers, then distills via MOHAWK on 12B tokens ($< 0.1\%$ of Llama's pre-training data). Llamba-8B scores 61.0 on MMLU (teacher: 68.0), 80.9 on PIQA (teacher: 81.1) — $\sim 99\%$ average quality across 8 benchmarks, with higher throughput and no KV cache

Part 5: RL for Mathematical Reasoning

- In Part 3, the reward signal came from a *learned* reward model (which can be hacked). What if we had a *perfect* reward signal? Mathematics provides exactly this: a proof is either correct or it is not, and a computer can check
- Mathematics is an **ideal domain** for RL-based training of LLMs:
 - ▶ Correctness is **objective**: a proof is either valid or it is not
 - ▶ Verification is **automated**: formal proof assistants (Lean, Coq, Isabelle) can check proofs without human involvement
 - ▶ The reward signal is **non-hackable**: the verifier implements the rules of logic, not a learned proxy
- Contrast with natural language tasks (e.g. summarisation, dialogue):
 - ▶ Quality is subjective; the reward model is a learned approximation of human preferences
 - ▶ Reward hacking is a persistent problem
- Can we train LLMs to generate *formal mathematical proofs* using RL, where the type checker provides a perfect reward signal?
- This part surveys recent work at the intersection of **LLMs**, **reinforcement learning**, and **formal mathematics**

Lean 4 as a Verification Environment

- **Lean 4** (de Moura & Ullrich, 2021) is a modern proof assistant based on the **Calculus of Inductive Constructions** (a dependent type theory)
- Key features relevant for RL:
 - ▶ **Tactics**: proofs are constructed interactively using *tactics* — commands that transform proof goals (e.g. `simp`, `ring`, `omega`, `linarith`)
 - ▶ **Mathlib**: a large, community-maintained library of formalised mathematics (>100,000 theorems covering algebra, analysis, topology, combinatorics, etc.)
 - ▶ **Compilation**: Lean compiles proofs and reports errors with precise diagnostic messages
- **Simple example**:

`theorem add_comm :  $\forall a b : \mathbb{N}, a + b = b + a$  := by omega`

- The tactic `omega` is a decision procedure for linear arithmetic over $\mathbb{N}$ and $\mathbb{Z}$. Lean verifies that `omega` solves the goal; if it does, the proof is accepted
- **For RL**: the LLM generates tactic sequences; Lean provides binary feedback (accepted/rejected)

AlphaProof (DeepMind, 2025)

- **AlphaProof** couples an **encoder-decoder Transformer** with AlphaZero-style RL to prove theorems in Lean 4
- **Training**: pre-train on code/maths data ($\sim 3 \times 10^{12}$ decoder tokens, $\sim 12 \times 10^{12}$ encoder tokens); fine-tune on Mathlib tactics; auto-formalise 1M natural-language statements into 80M Lean statements; then RL over verified proofs
- **Proof search via MCTS**: each node is a proof state, each edge a tactic application. When a tactic splits a goal into subgoals, a *product node* requires all branches to be proved. A value head estimates tactics remaining; values are back-propagated via the *hardest branch*
- **Test-time RL**: when faced with a hard target theorem T , AlphaProof does not attempt T directly. Instead, an LLM generates millions of *variations* of T — e.g. special cases ($n = 2$), weaker hypotheses, or simpler analogues. These variants range from easy to hard, forming a natural **curriculum**. AlphaProof attempts to prove each variant using MCTS; every successful proof is used to *retrain the policy on the fly* (i.e. at inference time, not during a separate training phase). As the policy improves on easier variants, it gradually becomes capable of proving harder ones, eventually reaching T itself. The reward is always binary (Lean accepts or rejects) — **no learned reward model** is involved
- **Results**: proved 3 of 5 non-geometry IMO 2024 problems (P1, P2, P6), achieving **silver-medal performance**. P6 was the hardest — only 5 of 609 human contestants earned full marks

DeepSeek-Prover-V2 (Xin et al., 2025)

- **DeepSeek-Prover-V2** applies GRPO (Part 3) to Lean 4 proof generation (7B and 671B parameters)
- **Cold start** — the key challenge is obtaining initial training data when the model cannot yet prove anything:
 - ① Prompt DeepSeek-V3 (a strong general-purpose LLM) to write a *natural-language proof sketch* of a hard theorem, and simultaneously translate it into a Lean skeleton where each intermediate step is left as a `sorry` placeholder
 - ② A smaller 7B prover model attempts to fill in each `sorry` independently — these are easier subproblems that it *can* solve even before RL training
 - ③ When all subgoals are resolved, the pieces are assembled into a complete formal proof. This proof, paired with DeepSeek-V3's reasoning chain, becomes an SFT training example

The resulting SFT model serves as π_{ref} for the GRPO phase

- **GRPO training**: the GRPO loss (2.13) with binary verifier reward ($R = 1$ if Lean accepts, 0 otherwise). Each iteration: 256 theorems $\times G = 32$ candidate proofs (max 32,768 tokens)
- **Results** (Pass@K): miniF2F-test **88.9%** (671B); PutnamBench **49/658** (671B); AIME 2024–25 **6/15** (671B)

Aristotle (Harmonic, 2025)

- **Aristotle** integrates three components: a Lean proof search system, an informal reasoning module, and a geometry solver. A single >200B-parameter Transformer serves as both policy and value function
- **Monte Carlo Graph Search (MCGS)**: generalises MCTS by treating Lean proof states as nodes in a *hypergraph* (not a tree), since different tactic sequences can reach the same proof state. Uses AND/OR structure: a state is proved if *any* action succeeds (OR); an action succeeds only if *all* resulting subgoals are proved (AND). Node selection uses a PUCT variant with an exploration bonus weighted by the policy prior
- **Informal reasoning**: for hard problems, an LLM generates a natural-language proof, restructures it as a sequence of lemmas with short individual proofs, then auto-formalises each lemma into Lean. Lean compiler feedback is used to iteratively correct formalisation errors. Proved lemmas are kept; failed ones are replaced with new decompositions
- **Test-time training**: alternate between (i) attempting the target problem via MCGS, and (ii) retraining the model on search traces from those attempts — specialising the policy to the problem at hand
- **Results**: solved 5 of 6 IMO 2025 problems with formally verified Lean proofs (**gold-medal equivalent**); only P6 remained unsolved

RL Training vs Iterative Refinement with Lean Feedback

- Two paradigms for LLM-based theorem proving in Lean:
 - ① **RL training**: train a specialised model with Lean's binary reward (AlphaProof, DeepSeek-Prover-V2, Aristotle)
 - ② **Iterative refinement**: prompt a frontier model, compile with Lean, feed diagnostics back, and repeat — **no RL training at all**
- **The case for RL**: the model internalises proof search strategies and tactic selection through millions of training episodes; test-time RL (AlphaProof) and MCTS/MCGS enable systematic exploration of the proof space
- **The case against RL**: frontier models improve rapidly; a compile-fix loop with Lean feedback is simple, requires no training infrastructure, and automatically benefits from each new model generation. RL-trained specialists risk being overtaken by the next frontier release

Summary

- **Pre-training** provides a capable base model; **SFT** teaches instruction-following; **LoRA** makes fine-tuning parameter-efficient
- The **RL formulation** of text generation enables reward-driven optimisation via policy gradients and advantage estimation (GAE)
- **PPO**: stable updates via clipping the importance ratio; requires a critic network
- **GRPO**: eliminates the critic via group-normalised advantages; used in DeepSeek-R1
- **RLHF**: reward model training (Bradley–Terry) + RL optimisation (PPO or GRPO)
- **Knowledge distillation**: DeepSeek-R1's 14B distilled model surpasses o1-mini on AIME via SFT on chain-of-thought traces; distillation into SSMs (Llamba) offers Transformer quality at sub-quadratic cost
- **Formal theorem proving**: the Lean type checker provides a non-hackable binary reward; AlphaProof (MCTS + test-time RL), DeepSeek-Prover-V2 (subgoal decomposition + GRPO), and Aristotle (MCGS + informal reasoning) achieve IMO medal-level performance